

1. 2012Q1

Which of the following tissue types has the greatest number of mitochondria?

- A. the wall of an air sac
- B. the wall of a capillary
- C. the inner wall of a lymph vessel
- D. the inner wall of the small intestine

2. 2012Q21

A red blood cell leaves the aorta, travels through the body and arrives at the capillaries of the air sacs. The correct sequence of organs through which the red blood cell may have travelled is:

- A. liver, lungs, small intestine and heart.
- B. lungs, heart, small intestine and liver.
- C. small intestine, heart, liver and lungs.
- D. small intestine, liver, heart and lungs.

3. 2012Q31

Questions 31 and 32 refer to the photomicrograph below, which shows some blood vessels found in the human body:



Blood vessel X is an artery because it has a

- A. thick muscular wall to withstand high blood pressure.
- B. thick muscular wall to generate high blood pressure.
- C. large lumen to slow down the blood flow.
- D. large lumen to accommodate more blood.

4. 2012Q32

If blood vessel X is connected with the lung, it carries

- A. oxygenated blood towards the heart.
- B. oxygenated blood towards the lungs.
- C. deoxygenated blood towards the heart.
- D. deoxygenated blood towards the lungs.

5. 2012Q33

In the circulatory system, the highest blood pressure is developed in the

- A. left atrium.
- B. right atrium.
- C. left ventricle.
- D. right ventricle.

6. 2013Q31

Muscles are found in

- (1) arterioles.
- (2) capillaries.
- (3) veins.
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

7. 2015Q14

As the blood in the capillary continues to flow, the red blood cell will first return to the

- A. left atrium.
- B. right atrium.
- C. left ventricle.
- D. right ventricle.

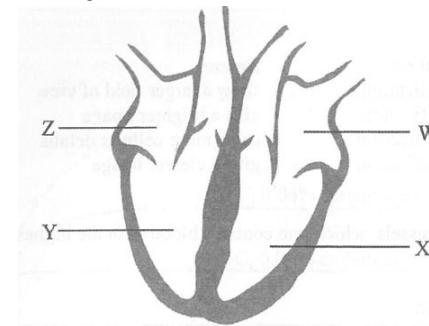
8. 2016Q13

Of the following blood vessels, which one contains blood with the highest urea content?

- A. renal vein
- B. hepatic vein
- C. pulmonary vein
- D. hepatic portal vein

9. 2016Q16

Questions 16 and 17 refer to the diagram below, which shows a section of the heart:



When W contracts, which of the following combinations about the conditions of the valves is correct?

- | | Semi-lunar valve | Bicuspid valve |
|----|-------------------------|-----------------------|
| A. | open | open |
| B. | open | closed |
| C. | closed | open |
| D. | closed | closed |

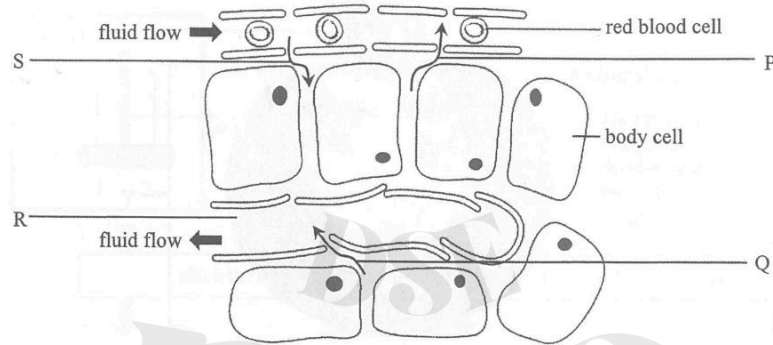
10. 2016Q17

Which of the following statements about the heart is correct?

- A. Z receives oxygenated blood from the lungs.
- B. Y and Z pump out same volume of blood.
- C. Blood in Z has higher glucose content than that of W.
- D. Blood pumped out from Y travels a longer distance than that of X before returning to the heart.

11. 2016Q28

Questions 28 and 29 refer to the diagram below, which shows the cells and the associated vessels in a tissue:



Which of the following combinations correctly identifies the major causes of fluid flow represented by P, Q and S?

- | | P | Q | S |
|----|----------------------|----------------------|----------------------|
| A. | osmosis | diffusion | hydrostatic pressure |
| B. | hydrostatic pressure | hydrostatic pressure | diffusion |
| C. | osmosis | hydrostatic pressure | hydrostatic pressure |
| D. | diffusion | diffusion | osmosis |

12. 2016Q29

Fluid in R eventually returns to the heart at the

- A. left atrium.
- B. right atrium.
- C. left ventricle.
- D. right ventricle.

13. 2017Q18

Which of the following descriptions of the circulatory system is correct?

- A. Veins have a large lumen because the blood pressure inside them is low.
- B. Arteries have a thick layer of elastic tissue because they have smaller lumens.
- C. The aorta has the highest blood pressure because it supplies blood to the whole body.
- D. The capillary network is a suitable site for material exchange because capillaries have the thinnest walls.

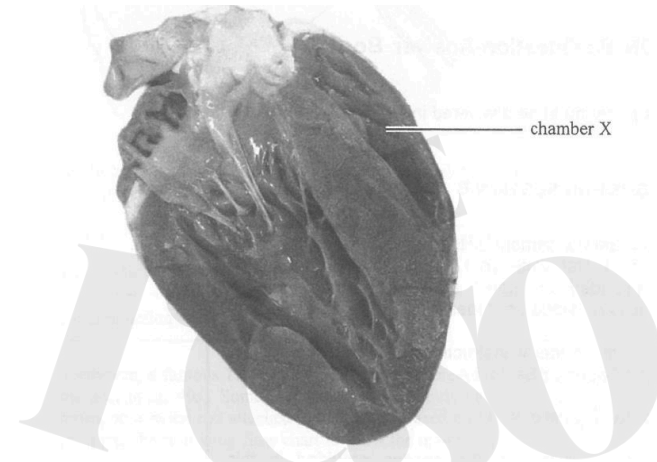
14. 2018Q34

Which of the following contribute to the continuous blood flow in the aorta?

- (1) pumping action of the heart
 - (2) elastic nature of the wall of the aorta
 - (3) contraction and relaxation of muscle wall of the aorta
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

15. 2018Q35

The following photograph shows a dissected pig heart:



Which of the following descriptions about chamber X is correct?

- A. It receives blood from the pulmonary vein.
- B. It pumps out blood to the aorta.
- C. It receives blood from the vena cava.
- D. It pumps out blood to the pulmonary artery.

16. 2019Q15

Which of the following description(s) of human red blood cells is/ are correct?

- (1) They cannot synthesise enzymes or proteins for repair because of the absence of the nucleus.
 - (2) They will stop functioning one day because haemoglobin will be used up.
 - (3) They do not have an energy supply because of the absence of mitochondria.
- A. (1) only
 - B. (2) only
 - C. (1) and (3) only
 - D. (2) and (3) only

17. 2019Q22

When ventricles contract, the valves between ventricles and atria close. Which of the following is the cause of the valve closure?

- A. The heart tendons hold the valves in position.
- B. The refilling of blood at the atria pushes the valves so that they close.
- C. The closure of valves prevents the blood from flowing back to the atria.
- D. The higher blood pressure resulting from ventricular contraction pushes the valves so that they close.

18. 2020Q12

Which of the following descriptions of the hepatic portal vein is correct?

- A. It carries blood away from the liver.
- B. It carries blood with lower oxygen content than that in the hepatic vein.
- C. It carries blood with lower amino acid content than that in the hepatic artery after meals.
- D. It carries blood with lower glucose content than that in the hepatic vein when one is hungry.

19. 2020Q14

Which of the following combinations correctly matches the water movement in the human body with its major driving force?

- | | Water movement | Major driving force |
|----|--|----------------------------|
| A. | water in tissue fluid enters lymph vessels | osmosis |
| B. | water in blood leaves capillaries at the arterial end | hydrostatic pressure |
| C. | water in tissue fluid enters capillaries at the venous end | active transport |
| D. | water enters capillaries from the lumen of the small intestine | diffusion |

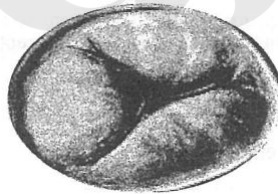
20. 2021Q20

Carbon dioxide produced in the small intestine is passed to the lungs for gas exchange. Which of the following correctly shows the transport route of the carbon dioxide?

- A. small intestine → lungs
- B. small intestine → liver → lungs
- C. small intestine → heart → lungs
- D. small intestine → liver → heart → lungs

21. 2022Q19

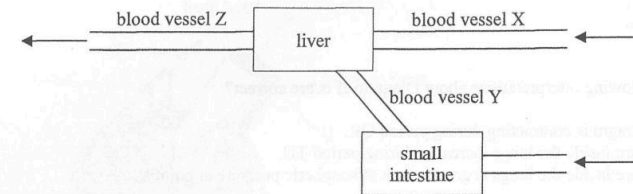
Imagine that you were a red blood cell travelling along a blood vessel and had reached the heart. After passing through the 'gate' shown in the diagram below, there was another 'gate' ahead. Which heart chamber were you situated now?



- A. left atrium
- B. left ventricle
- C. right atrium
- D. right ventricle

22. 2022Q21

Questions 21 and 22 refer to the diagram below, which shows part of the human circulatory system and the associated organs:



Which of the following correctly compares the blood glucose concentration in blood vessels X, Y and Z when a person is fasting?

- A. $X > Z > Y$
- B. $Y > X > Z$
- C. $Z > Y > X$
- D. $Z > X > Y$

23. 2022Q22

Which of the following descriptions about blood vessel Y are correct?

- (1) It transports hormones.
 - (2) It carries deoxygenated blood.
 - (3) It has capillary networks at both ends.
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

LQ

1. 2015Q11 - Essay

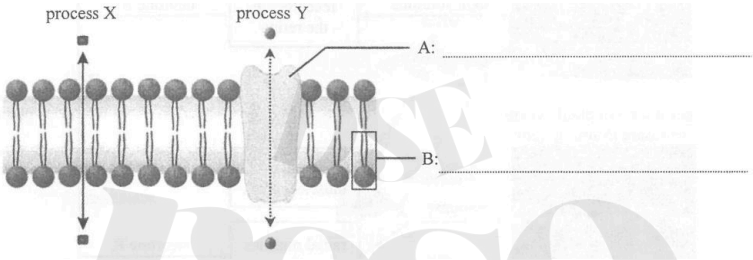
Although both arteries and veins are blood vessels, they are very different in their structure. Discuss how their structural differences are related to the different ways of maintaining blood flow inside the blood vessels. (11 marks)

2. 2017Q11 - Essay

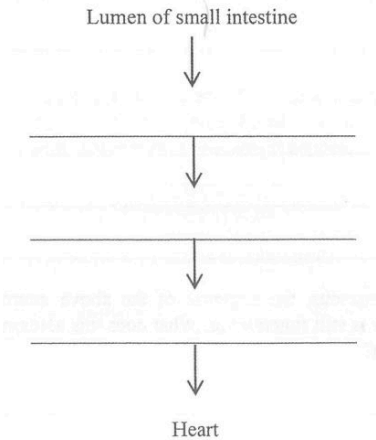
Some natural therapists claim that applying pressure along one’s limbs toward the body trunk can improve the circulation of lymph and result in reduced body weight. However, the effects of this treatment are controversial. Briefly describe how lymph is formed from blood and returned to the blood circulatory system. For each of the claims above, discuss whether it is scientifically valid. (11 marks)

3. 2020Q4biii

The following diagram shows a cell membrane and the movement of some substances across it:



- b) X and Y represent two different processes by which substances pass through the cell membrane.
- iii) The digested products absorbed through process X are transported from the small intestine to the heart. Complete the following flowchart to show the vessels involved in the transport. (3 marks)

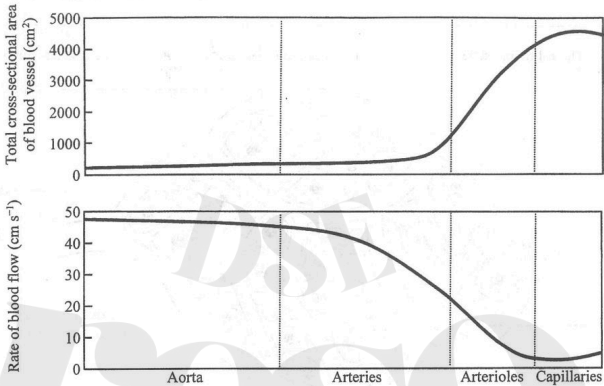


4. 2020Q11 - Essay

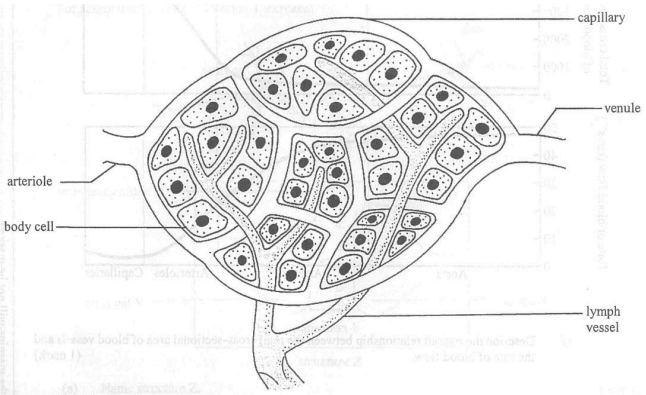
Our body systems are highly specialised at various levels of organisation to facilitate the functioning of each system. Discuss how the adaptive features of the human blood and circulatory system facilitate the transport of oxygen. Your discussion should cover adaptive features at cellular, tissue, and organ levels. (11 marks) (Note: Details of the different types of blood vessels are not required.)

5. 2021Q3

- a) The graph below shows the total cross-sectional area and the rate of blood flow of different types of blood vessels:



- i) Describe the overall relationship between the total cross-sectional area of blood vessels and the rate of blood flow. (1 mark)
- ii) Explain how the relationship described in (i) can facilitate the material exchange that takes place in the capillaries. (2 marks)
- b) The following schematic diagram illustrates a capillary network and the associated structures:



With reference to **two** features of the capillary network illustrated in the above diagram, explain the importance of these features to the material exchange in the capillary network. (4 marks)

Features illustrated in the diagram	Importance to the material exchange

6. 2023Q10bii3

Cassava is a crop which grows in areas with poor soil and a low rainfall. It produces starchy root tubers which serve as a major food source in Africa.

- b) Table I below shows some nutritional information of cassava while Table II lists the daily energy and protein requirements recommended for boys at age 16:

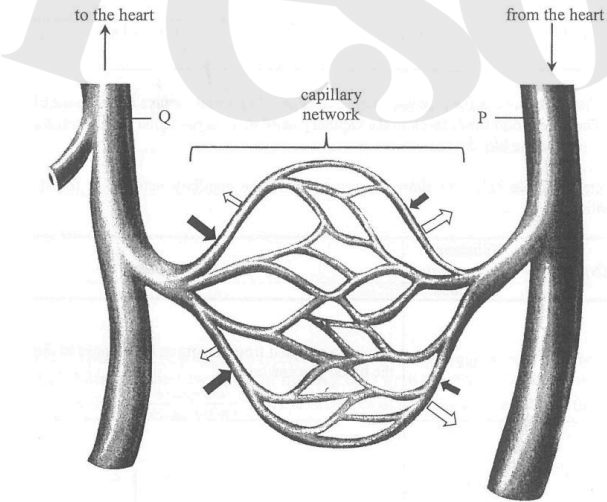
Table I		Table II	
Fresh weight (g) from which 100 g dry weight is yielded	250		Daily requirement
Energy (kJ per 100 g dry weight)	2 675	Energy (kJ)	11 100
Protein (g per 100 g dry weight)	3.5	Protein (g)	52

In Africa, some low-income families may rely only on cassava for food for a long period.

- i) A 16-year-old boy relies only on cassava for food. Calculate the fresh weight of cassava he needs to consume so as to meet the recommended daily energy requirement. (1 mark)
- ii) After consuming cassava only for a period of time, this boy develops swollen feet due to the accumulation of tissue fluid.
- 1) How much protein can be obtained from the amount of cassava consumed in a (i)? (1 mark)
- 2) According to Table II, predict the difference of the blood protein level of this boy when compared with that of normal healthy boys of the same age. Explain your answer. (2 mark)
- 3) Based on your answer in (2), explain why this would lead to the accumulation of tissue fluid in his feet. (2 marks)

7. 2024Q4a, b

The schematic diagram below shows the arrangements of some blood vessels:



- a) The two types of arrows (black and white) represent two factors which govern the movement of fluid into or out of the capillary network. Identify these two factors. (2 marks)
- Black arrow:
- White arrow:
- b) The sizes of these arrows in the above diagram represent the magnitudes of the factors. Explain the change in the factors denoted by (white arrow) as blood flows from P to Q. (3 marks)

Ans

2012Q1: D (73%)	2012Q21: D (83%)	2012Q31: A (90%)	2012Q32: D (75%)
2012Q33: C (76%)	2013Q31: B (64%)	2015Q14: A (67%)	2016Q13: B (47%)
2016Q16: C (76%)	2016Q17: C (40%)	2016Q28: C (18%)	2016Q29: B (73%)
2017Q18: D (49%)	2018Q34: A (38%)	2018Q35: D (20%)	2019Q15: A (35%)
2019Q22: D (39%)	2020Q12: D (59%)	2020Q14: B (53%)	2021Q20: D (70%)
2022Q19: D (51%)	2022Q21: D (47%)	2022Q22: D (27%)	

2015Q11

11	Structural differences (S: max. 3)	Ways of maintaining blood flow (F: max. 5)	
	- wall of arteries is thicker than that of veins (1)	- Pumping of heart created a high blood pressure to drive the blood flow in arteries (1), the thick wall can withstand the high blood pressure (1)	(1+2)
	- wall of arteries contains more elastic tissue than that of veins (1)	- Elastic nature of the arterial wall allows recoil of the wall (1) which maintains the blood flow along the arteries	(1+1)
	- valves are present in veins but not the arteries (1)	- blood flow in veins is maintained by the contraction of adjacent skeletal muscles (1), which squeeze the blood along, presence of valves can prevent the back flow of blood (1)	(1+2)
	- lumen of veins is larger than that of arteries (1)	- Blood pressure inside veins is relatively low (1), having a larger lumen reduces the resistance to blood flow / to increase the rate / amount of blood flow (1)	(1+2)

2017Q11

- 11 Formation of lymph and its return (max. 5)
- blood pressure / hydrostatic pressure at the arteriole end of the capillary bed is higher than that of the tissue fluid, this forces some of the plasma out (1)
 - the remaining blood cells / proteins are left in the blood and drain to the venous end (1)
 - water moves into the capillary bed at the venous end by osmosis (1)
 - excess tissue fluid will enter the lymph vessels to form the lymph due to the higher hydrostatic pressure (1) in the tissue spaces
 - with the assistance of the contraction of skeletal muscles to squeeze the lymph vessels (1)
 - and the presence of valves in lymph vessels to prevent the back flow of lymph (1)
 - thus, the lymph will flow along the lymph vessels and eventually return to the heart (1)

whether the claims are scientifically valid

Improvement of Circulation (max. 1)

Valid:

- as the valves of the lymph vessels prevent back flow, pressing along the lymph vessels, especially along the limbs towards the trunk, does help the return of lymph to the heart (1)

OR

Not valid:

lymph vessels are usually found deep inside the body (1) which makes it difficult to improve the circulation by pressing

Reduction in Body weight (max. 2)

Not valid:

- this may reduce the accumulation of tissue fluid / lymph at the limbs (1), giving a slimmer appearance temporarily
- the lymph only returns to the circulatory system but is not eliminated (1), i.e., no change in body weight
- this does not affect the energy input / energy output / food consumption of the body (1)
- therefore, there should be no effect on body weight (1)

Effective communication (0-3)

(max. 1)

(max. 3)
11 marks

2020Q4biii

- 4 (b) (iii) lumen of small intestine
- ↓
- lacteal (1)
- ↓
- lymph vessel (1)
- ↓
- subclavian vein / vena cava (1)
- ↓
- heart

2020Q11

- 11 Cellular level:
- red blood cells are biconcave disc in shape (1), which increases surface area to volume ratio for faster gas exchange (1)
 - red blood cells are equipped with hemoglobin (1), which is specialized for binding with oxygen (1)
 - in red blood cells nucleus is absent / membrane-bound organelles are absent (1), leaving more space for hemoglobin (1)

max. 4

Tissue level:

- blood is a fluid or liquid tissue carrying the red blood cells / low viscosity allow flowing of the blood (1)

Organ / system level:

- heart is made up of cardiac muscles (1), which give driving force to maintain blood flow (1)
- septum or wall divides the heart into two halves (1), with one half for carrying deoxygenated blood to the lungs (1), and the other half for carrying oxygenated blood to other organs / body parts (1), increasing the efficiency of the oxygen transport
- blood vessels with lumen act as channels for distribution of blood to various organs (1)

max. 4

Effective communication (C)max. 3
11 marks

2021Q3

- 3 (a) (i) • the greater the total cross-sectional area, the slower the blood flow / the smaller the total cross-sectional area, the faster the blood flow (1)
- (ii) • the blood flow in the capillaries is the slowest / very slow (1)
- this allows more / enough / sufficient time for the exchange of materials in the capillaries (1)

(1)

(2)

(b)	Features illustrated in the diagram		Importance to material exchange	
	the capillaries are highly branched / a dense network (1)		to increase the surface area for the exchange of materials (1)	
	the capillaries penetrate tissues / reach most of the cells in the tissues (1)		to shorten the diffusion distance (1)	

(4)

7 marks

2023Q10bii3

- 10 (b) (i) $11100 / 2675 \times 250 = 1037.5 \text{ g (1)}$
[no unit no mark]
- (ii) (1) $11100 / 2675 \times 3.5 = 14.5 \text{ g (1)}$
[no unit no mark]
- (2) • his blood protein level will be lower than normal people (1)
- because his protein intake is much lower than the recommended daily requirement (1)
- (3) • the low blood protein level leads to a higher water potential than normal at the venous end of the capillary network (1)
- the amount of water in tissue fluid returning to the blood at the venous end of the capillary will be reduced (1)
- hence, tissue fluid will accumulate

(1)

(1)

(2)

(2)

2024Q4a, b

	Concept for mark award	Example	Marks
4(a)	correct identification of two factors	black arrow: difference between the water potential of the blood and tissue fluid / the water potential gradient between blood and tissue fluid	(1)
		white arrow: difference between the hydrostatic pressure of blood and tissue fluid	(1)
4(b)	change in the hydrostatic pressure	the factor represented by white arrow / hydrostatic pressure decreases	(1)
	fluid movement	as fluid / water is forced out of the blood at the end near P	(1)
	consequence for blood volume	the blood volume in the capillaries decreases, leading to the decrease in hydrostatic pressure / blood pressure	(1)

DSE
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